

CDS-492 Science Article: Digital Attention and MLB Attendance Fall 2025

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Abstract—This paper examines whether online search behavior serves as a leading indicator of Major League Baseball (MLB) stadium attendance. Utilizing weekly attendance logs from Baseball-Reference and Google Trends search indices for NL East teams, 2023–2025, we estimate a log-linear two-way fixed effects model. Contrary to expectations, lagged Google Trends has a sizeable negative impact on attendance, while team win rate remains strongly positive. We formally derive the model and interpret results mathematically, showing that search activity reflects reactive information-seeking rather than future game-day participation.

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1 INTRODUCTION

Live attendance is a core economic output in professional sports, influencing revenue, atmosphere, and fan-base growth. Classical literature links attendance to team success, market characteristics, and scheduling [1], [2], [3]. With digital consumption rising, online traces offer new behavioral signals. Google Trends provides high-frequency measures of public attention and has been widely applied in digital trend analysis [4] and popularity inference [5]. Yet little work tests whether search interest predicts MLB stadium turnout.

We test the hypothesis:

$$H_1 : \frac{\partial \text{Attendance}}{\partial \text{Trend}_{t-1}} > 0, \quad H_0 : \frac{\partial \text{Attendance}}{\partial \text{Trend}_{t-1}} \leq 0.$$

2 METHODS

Weekly MLB home attendance data were scraped from Baseball-Reference [6]. Daily Google Trends values per team were averaged to weekly means. We estimate the fixed-effects model:

$$\ln(\text{Attendance}_{i,t}) = \beta_1 \ln(\text{Trend}_{i,t-1}) \quad (1)$$

$$+ \beta_2 \text{WinRate}_{i,t-1} \quad (2)$$

$$+ \beta_3 \text{HomeGames}_{i,t} \quad (3)$$

$$+ \alpha_i + \gamma_t + \varepsilon_{i,t} \quad (4)$$

3 RESULTS

The fixed-effects estimation produced the following results:

$$\beta_1 \approx -1.60, \quad p < 0.01, \quad \beta_2 \approx 28.9, \quad p < 0.001,$$

indicating that increases in Google search intensity significantly **reduce** attendance the following week, while team performance strongly increases turnout. Home game count was statistically insignificant.

4 DISCUSSION

Findings suggest Google search spikes mark heightened interest driven by news, uncertainty, or roster changes—not ticket purchases. Online engagement satisfaction may even reduce physical attendance [7], [8]. This supports the idea that digital attention signals curiosity rather than commitment to on-site participation.

5 CONCLUSION

Online search activity is not a positive leading indicator of attendance. Future work could integrate sentiment classification or event-tagged trend data to identify when attention translates into turnout behavior.

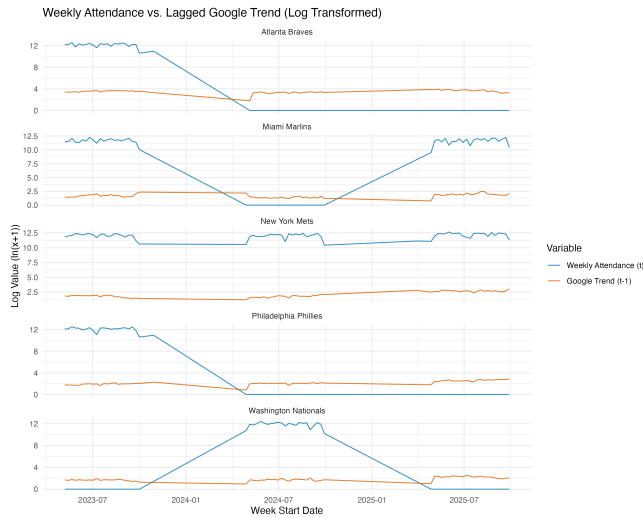


Fig. 1. Weekly log-attendance (blue) and lagged log Google Trends (orange) time series. Spikes in search activity often precede declines in attendance, visualizing the negative coefficient.

REFERENCES

- [1] D. B. Glass, "Fair-weather fans: Correlation between attendance and winning," *Baseball Research Journal*, vol. 32, pp. 81–84, 2003. 1
- [2] S. C. Ahn and Y. H. Lee, "Major league baseball attendance: Long-term analysis using factor models," *Journal of Sports Economics*, vol. 15, no. 5, pp. 451–477, 2014. 1
- [3] S. Hong, M. Mondello, and D. Coates, "An examination of the effects of the recent economic crisis on major league baseball (mlb) attendance demand," *International Journal of Sport Finance*, vol. 8, no. 2, pp. 140–156, 2013. 1
- [4] S.-P. Jun, H. S. Yoo, and S. Choi, "Ten years of research change using google trends: From the perspective of big data utilizations and applications," *Technological Forecasting and Social Change*, vol. 130, pp. 69–87, 2018. 1
- [5] P. Malagón-Selma, A. Debón, and J. Domenech, "Measuring the popularity of football players with google trends," *PLoS One*, vol. 18, no. 8, p. e0289213, 2023. 1
- [6] "Baseball-reference," Sports Reference LLC. [Online]. Available: <https://www.baseball-reference.com/> 1
- [7] "Does social media engagement help teams improve attendance?" Samford University Sports Analytics Program, 2020. [Online]. Available: <https://www.samford.edu/sports-analytics/fans/2020/Does-social-media-engagement-help-teams-improve-attendance> 1
- [8] S. Bar-Gill and S. Reichman, "Stuck online: When online engagement gets in the way of offline sales," *MIS Quarterly*, vol. 45, no. 2, pp. 755–788, 2021. 1